

**Department of Computer Science and Engineering
BRAC University**

Set A

Examination: Semester Final
Duration: 1 hour 50 minutes

Semester: Fall 2025
Full Marks: 40

CSE 423: Computer Graphics

Name:	ID:	Section:
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*Answer the following questions. Understanding the questions is part of the examination.
The numbers on the right margin indicate marks.*

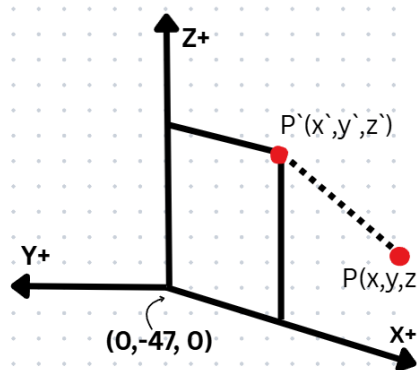
Question 1 [CO3]

- a. **Briefly explain** why the RGB color model cannot be directly used for physical printing works. **Suggest** an alternative model that would be appropriate for this task and **justify** your choice. **2+2**
- b. Standard displays use an 8-bit per channel system, meaning each RGB channel is allocated 8 bits to represent intensity levels (0 – 255). In contrast, modern HDR10 displays utilize a **10-bit per channel** system. **Calculate** the total number of unique RGB color combinations a 10-bit per channel system can represent. An HDR10 pixel is recorded with the raw **10-bit RGB** values of **(921, 153, 512)**. **Convert** this pixel into the **HSV color space**. **2+4**

Question 2 [CO1]

- a. Thorfinn observes a spherical object illuminated by a **flashlight** from **one direction**. He notices that only the side facing the flashlight is visible, while the opposite side remains completely **dark**. Using **Phong's Reflection Model**, **explain** why this phenomenon occurs. He further observes a **broad, well-distributed highlight** on the illuminated region of the sphere. **Suggest** a **minimal** value for the **shininess exponent** of the object and justify your answer. **2+2**
- b. Another sphere has a center at **C(1, 2, 3)**. The sphere is illuminated by **two** point light sources, **L₁** and **L₂**, with an **intensity of 5** each, located at **(4, 5, 6)** and **(8, 10, 12)**, respectively. Each light source has a radius of influence of **30 units**. At the surface point of illumination **P(10, 6, 5)**, Thorfinn notices that the **reflected light** from both sources travels **directly toward his eyes**. The scenario has a **constant ambient intensity vector** of **2**. The material of the sphere has an **ambient coefficient vector** of **<0.3, 0.2, 0.7>**, a **diffuse coefficient vector** of **<0.4, 0.1, 0.3>**, and a **specular coefficient** of **0.5**, a shininess exponent of **10**. Now, **compute the total illumination intensity** $I = \langle I_r, I_g, I_b \rangle$ at the surface point **P**, considering only the light source attenuation. **6**

Question 3 [CO2]



- a. Yuta projected a point $P(x, y, z)$ according to the above diagram and got the point $P'(x', y', z')$. **Derive the 4x4 homogenous matrix** for the given scenario and **express** it in terms of the Inclination Angle and the Angle of Skewness. 5
- b. A perspective projection is defined by a **center of projection (COP)** and a **projection plane (PP)**. The center of projection is located at $(33, 11, 23)$. A point in 3D space is given by $P(10, 55, 675)$. The projection plane is the **XY-plane**. It is located at a distance of **400** units away from the COP and is perpendicular to the viewing direction. **Sketch** a labeled diagram of the setup and **compute** the coordinates of the projected point, P' , defined as the intersection of the projection ray with the projection plane. 1+4

Question 4 [CO3]

- a. When constructing curves using the *Lagrange polynomial*, **what** type of representation is being used? Briefly **explain** why these representations are generally avoided when continuity and smoothness must be guaranteed in computer graphics applications. 4
- b. Two parametric splines are defined as $C(t) = (x(t), y(t)); [0 \leq t \leq 1]$ where the second spline is joined at the endpoint of the first spline ($q = 1$) to form a curve. 2+4

Spline 1: $C_1(q) = (q^2, 2q), 0 \leq q \leq 1$

Spline 2: $C_2(s) = (1 + 2s + ks^2, 2 + 2s + \frac{1}{2}ks^2), 0 \leq s \leq 1$

[Here, k is an arbitrary value]

- i. **Determine** the value(s) of k for which the composite curve is C^2 **continuous** at the joint.
- ii. **Determine** the value(s) of k for which the composite curve is **only G^2 continuous and for which it is not**. Briefly **explain in two sentences** whether G^2 continuity can exist without C^2 continuity, referring explicitly to tangent direction vs. magnitude.